

B-INFOHUB BRIDGE v1.0 -- Briggs & Stratton GC-1032 RS-485 bridge with InfoHub passthrough

POWER IN: 12 V from the generator harness. Q1 reverse-polarity (carries the InfoHub passthrough too, 4 A part). D2 TVS clamps starter load-dump. F2 fuses the InfoHub separately.

ESP32-DevKitC-38, socketed. Use the -32U variant for an external antenna. No strapping pin (0/2/12/15) is used, so nothing here can stop it booting.

5 V RAIL: AP63205WU-7 sync buck, 3.8-32 V in, fixed 5 V 2 A (from b-hydro carrier v2.1)

power flags

PORT A -- GENERATOR CONTROLLER. We are Modbus MASTER here. Slave 10, 9600 8E1 (measured). RE and DE tied: one GPIO (IO4) drives direction. JP1 fits the 120R terminator.

STATUS LEDs: PWR and TX need no GPIO -- TX is driven from DE, so it proves the bridge is actually transmitting, not merely powered.

PORT B -- INFOHUB PASSTHROUGH. We are Modbus SERVER here, impersonating the controller at slave 10 so the InfoHub keeps working (ESPHome modbus role: server + modbus_server).

EXTERNAL LED HEADER: the on-board LEDs are invisible inside a sealed box. J4 carries the same anode nodes so panel LEDs can sit in parallel.

mounting

b-infohub

Sheet: /
File: binfohub-bridge.kicad_pcb

H1

M1

H2

M2

H3

M3

H4

M3

H5

M3

H6

M3

Title: B-Infohub Bridge v1.0

Size: A3

Date: 2026-08-31

KiCad E.D.A. 10.0.5

Rev: 1.0

Id: 1/1